

Module 6: Particles and medical physics

In this module, learners will learn about capacitors, electric field, electromagnetism, nuclear physics, particle physics and medical imaging.

6.1 Capacitors

This section introduces the basic properties of capacitors and how they are used in electrical circuits. The use of capacitors as a source of electrical energy is then developed. This section introduces the mathematics of exponential decay, which is also required for the decay of radioactive nuclei in **6.4**.

This section provides knowledge and understanding of capacitors and exponential decay.

Experimental work provides an excellent way to understand the behaviour of capacitors in electrical circuits and the management of safety and risks when using power supplies (HSW4). There are many opportunities for learners to use spreadsheets in the analysis and presentation of data (HSW3). The varied uses of capacitors give the opportunity for the consideration of their use in many practical applications (HSW2, 5, 6, 9)

6.1.1 Capacitors

Learning outcomes		Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>		
(a)	capacitance; $C = \frac{Q}{V}$; the unit farad	
(b)	charging and discharging of a capacitor or capacitor plates with reference to the flow of electrons	HSW2
(c)	total capacitance of two or more capacitors in series; $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$	
(d)	total capacitance of two or more capacitors in parallel; $C = C_1 + C_2 + \dots$	
(e)	(i) analysis of circuits containing capacitors, including resistors	HSW5
	(ii) techniques and procedures used to investigate capacitors in both series and parallel combinations using ammeters and voltmeters.	PAG9